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CONTEXT SHIELD

- **Problem Statement ID-** SIH26171
- **Problem Statement Title-** On-Device Visual Perception for Light-weight Browser Agents
- **Theme-**Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name:** Chr0m1um

CONTEXT SHIELD

Browser assistance with local perception and explicit control

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THE PROBLEM

Useful context can expose private information.

Raw screens may contain identities and credentials. Untrusted page content can also redirect an agent's actions [5,6].

OUR APPROACH

Read the screen locally.
Send sanitized context for planning.
Check permission before private actions.

Every private fill needs one-use approval for an exact destination and field.

CAPTURED PLANNER REQUESTS

SYNTHETIC LOCAL DATA	OBSERVED OUTBOUND BODY
privacy.fixture@example.com	Original string absent
Password: NeverSendThis!	Original string absent

LOCAL PORTRAIT



OUTGOING IMAGE CROP



9 planner requests across 3 runs. Known test strings absent.
Planning traffic only. Approved webpages can still read filled values.

TECHNICAL APPROACH

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ON-DEVICE TRUST ZONE

1 CAPTURE

DOM, accessibility data
and visible pixels

2 PERCEIVE

YOLOX faces [1]
PP-OCR text [2]
Rampart contextual PII

3 PROTECT

Mask detected secrets.
Use LOCAL_* references.
Keep raw values locally.

OUTBOUND: sanitized task, safe elements, handles and redacted image crops

SERVER PLANNER

4 REASON [3]

Qwen reasons over safe
text and redacted pixels.
Returns one typed action.

Qwen3-VL

RETURN: proposed action only

5 AUTHORIZE [7]

Check field, origin and page.
Require one-use consent
before private-value release.

6 EXECUTE

Resolve the approved value.
Fill only the authorized field.

7 VERIFY

Observe the page again.
Confirm the expected change.
Stop safely on mismatch.

The server cannot authorize a private fill.

8 AUDIT [8] Hash-link non-secret events and sign a checkpoint.

Chrome + Firefox	WXT + TypeScript	ONNX Runtime Web	YOLOX + PP-OCR	Rampart	FastAPI	Qwen3-VL
Browser extension	UI & content scripts	Local inference	Vision & OCR	Contextual PII	Strict gateway API	Open-weights planner

Signed local ledger: SHA-256 + ECDSA P-256. No credentials in blocks. Gate enforcement and audit are separate.

FEASIBILITY AND VIABILITY

Measured prototype results against the five SIH evaluation areas

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19 SEP 2026 Apple M4 / 24 GiB / Chromium / local Qwen3-VL-4B

PS AREA / WEIGHT	OBSERVED RESULT	SCOPE / REMAINING LIMIT
Visual context / 25%	3/6 canvas tasks complete	Values absent from DOM. Price cases stop on over-redaction.
PII detection / 20%	68.4% precision 100.0% recall	11 authored pages, case/type scoring. 13 correct, 6 extra, 0 missing types.
Redaction / 20%	4/4 crops cover annotations	Two canvas secrets + one face at two sizes. Masking also hides surrounding content.
Client resources / 20%	2.60 GiB active peak 0.88 GiB baseline	Whole Chromium process tree. Extension-only CPU / RAM still unisolated.
Task latency / 15%	52.74 s median 3/3 judge runs complete	Cold client, already-running Qwen server. Three runs cannot establish tail latency.

Controlled regression evidence, not an official SIH score or an independent unseen-site benchmark.

Next: reduce privacy false positives, measure on low-end devices, and validate on independently labelled unseen tasks.

IMPACTS & BENEFITS

Controlled railway outcomes and measured privacy evidence

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01 BOOKING COMPLETED

The packaged extension reached a simulated ticket in assisted and rehearsal tests.

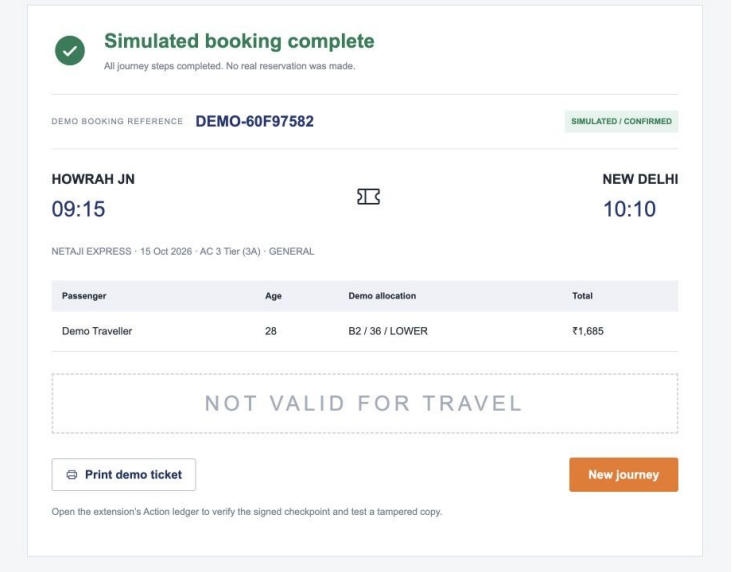
02 HOSTILE PRIVATE FILL BLOCKED

A malicious server action targeted an unrelated field. The field stayed empty and the gate recorded BLOCKED.

03 MODIFIED LOG REJECTED

An edited ledger copy failed integrity verification. A separately retained checkpoint anchors the check.

END-TO-END SANDBOX JOURNEY



Actual sandbox UI. Synthetic data and payment.
No IRCTC affiliation or live reservation.

19 Sep checks: 99 unit tests and 36 backend tests pass. Railway and malicious-fill regressions remain scoped tests.

Synthetic data and simulated payment. No live reservation or universal prompt-injection resistance claim.

RESEARCH AND REFERENCES

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PERCEPTION & BROWSER AGENTS

[1] Ge et al., 2021. Technical report (arXiv)
YOLOX: Exceeding YOLO Series in 2021

Detector design. The face checkpoint needs separate validation.

[2] Du et al., 2020. Technical report (arXiv)
PP-OCR: A Practical Ultra Lightweight OCR System

Lightweight OCR family. Deployed v6 files are cited in notes.

[3] Bai et al., 2025. Technical report (arXiv)
Qwen3-VL Technical Report

Multimodal reasoning background. No privacy guarantee.

[4] Koh et al., 2024. ACL
VisualWebArena: Evaluating Multimodal Agents on Realistic Visual Web Tasks
Evaluation of web tasks that need visual information.

PROMPT INJECTION, AUTHORIZATION & AUDIT

[5] Debenedetti et al., 2024. NeurIPS
AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents
Evaluate task success alongside resistance to attacks.

[6] Zhan et al., 2024. Findings of ACL
InjecAgent: Benchmarking Indirect Prompt Injections in Tool-Integrated Large Language Model Agents
Malicious external content and private-data exfiltration.

[7] Saltzer & Schroeder, 1975. Proceedings of the IEEE
The Protection of Information in Computer Systems

Least privilege and authorization checks on every release.

[8] Schneier & Kelsey, 1999. ACM TISSEC
Secure Audit Logs to Support Computer Forensics

Audit-log integrity. Our local ledger has narrower guarantees.

Research informs the design. ContextShield has not yet run benchmarks [4–6].

SOURCE CODE **RAILWAY VALIDATION** **RAILWAY DEMO GUIDE**

New measurements accompany this deck in FINAL-VALIDATION.md. Public links reference the published baseline.